

Rishon Ghosh

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Education

Simon Fraser University

Bachelor of Science, Computer Science (Joint Major with Business) | **GPA: 3.89**

Burnaby, BC, Canada

Expected Grad Date: Dec 2028

- **Coursework:** Algorithms & Data Structures, Introduction to Software Engineering, Business Technology Management
- **Awards:** Dean's Honour Roll, President's Honour Roll, SFU Undergraduate Open Scholarship

Experience

Software Engineer Consultant

Self-Employed, RG Tech

Vancouver, BC

Dec 2025 - Present

- Architected Python and Flask REST backends for **3 small-business clients**, turning existing sales and inventory data into reports
- Engineered SQL schemas and ETL scripts consolidating spreadsheet and POS data, saving **12 hours** of manual entry weekly
- Optimized database performance through indexing and query refactoring, reducing average API response times by **40%**
- Implemented secure authentication with password hashing and input validation to protect client and customer records
- Executed client discovery, scoping, and sprint planning to translate business needs into technical requirements delivered on schedule

Projects

TrustLine | *TypeScript, React, Node.js, ElevenLabs, Gemini, Twilio, Supabase, Docker*

Oct 2026

- Architected a real-time audio intelligence pipeline streaming live phone calls from Twilio Media Streams to ElevenLabs Scribe v2 Realtime over WebSockets, transcribing **8 kHz** telephony audio with no transcoding and warning callers mid-call in **5 languages**
- Engineered a dual-layer scam detection engine fusing a **~0.01 ms** rules layer across **12 scam tactics** with schema-constrained, Zod-validated LLM scoring (Google Gemini), achieving **100% precision and recall** with **0 false positives** on an 18-call labelled set at **1.1 s** median latency
- Established an evaluation and CI framework measuring precision, recall and alert latency, gating every push through GitHub Actions with **187 automated tests** across **28 suites**
- Hardened security and privacy with Twilio signature verification on webhooks and media streams, single-use transcription tokens, on-device PII redaction, consent-gated sharing and **10-minute** call caps
- Orchestrated production deployment on **\$0** infrastructure: a Dockerized call server on Railway with health-checked, path-filtered auto-deploys, and a React/Vite frontend with serverless APIs on Vercel

FlashCart | *Go, TypeScript, React, AWS Lambda, API Gateway, DynamoDB, SQS, CDK*

Sep 2026

- Spearheaded a serverless flash-sale API in Go on AWS Lambda and API Gateway, load-tested with k6 at **1,000+** concurrent users
- Devised oversell-proof inventory reservations with DynamoDB conditional writes and idempotency keys, recording **0** oversold units
- Orchestrated asynchronous order fulfillment through SQS with dead-letter queue retries, holding checkout p99 latency under **150ms**
- Established infrastructure as code with AWS CDK in TypeScript and GitHub Actions CI/CD, deploying the full stack on every merge

Graphite Gambit | *Java, libGDX, JUnit, Mockito, Maven, JaCoCo*

Jan 2026

- Led a team building a 2D top-down game in Java and libGDX, establishing weekly agile sprints and a merge-conflict protocol
- Created a map-loading system and collision engine, using mathematical proofs in testing to verify safe entity movement and bounds
- Automated a testing suite with JUnit and Mockito, achieving **71%** branch coverage across core game packages

Rubik's Cube Solver | *Java*

Oct 2025

- Programmed an optimal Rubik's Cube solver in Java using Iterative Deepening A* (IDA*) search, finding solutions in under **10s**
- Constructed memory-efficient pattern databases using combinatorial indexing and factorial mapping for $O(1)$ heuristic lookups
- Compressed a **43-quintillion** state space into a **~2MB** footprint, keeping heuristic data fully in memory during search

Technical Skills

Languages: Go, C++, TypeScript, Java, Python, SQL, C, JavaScript, HTML/CSS

Frameworks & Libraries: React, Angular, gRPC, Protocol Buffers, Flask, libGDX, JUnit, Mockito, Google Test

Cloud & DevOps: AWS (Lambda, API Gateway, DynamoDB, SQS, S3, CloudFront, CloudWatch, CDK), Docker, GitHub Actions

Databases: DynamoDB, SQLite

Developer Tools: Git, CMake, Maven, k6, JaCoCo, ThreadSanitizer, VS Code

Concepts: Distributed Systems, Fault Tolerance, Consistent Hashing, Concurrency, REST APIs, CI/CD, Infrastructure as Code

Business: Financial Accounting, Excel, PowerPoint, Sprint Planning & Execution